Literature

Wiki *Enhanced dissipation for passive scalars*

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1 Reviews

- A. L. Mazzucato, Y. Feng and C. Nobili, *Enhanced dissipation by advection and applications to PDEs*, Applied Mathematical Modelling **150** (2026), Part A, 116310, arXiv:2501.17695.
- M. Coti Zelati, G. Crippa, G. Iyer and A. L. Mazzucato, *Mixing in incompressible flows: transport, dissipation, and their interplay*, Notices of the American Mathematical Society **71** (2024), 593–604, arXiv:2308.00358.

2 Deterministic enhanced dissipation

Shear, radial and other symmetric flows.

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- J. Benthaus, G. M. Coclite and C. Nobili, *Mixing and enhanced dissipation in a time-translating shear flow*, preprint (2026), arXiv:2603.14624.
- T. Li and L. Zhang, *Enhanced dissipation and Taylor dispersion by a parallel shear flow in an infinite cylinder with unbounded cross section*, preprint (2025), arXiv:2510.13097.
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- Y. Feng and A. L. Mazzucato, *Dissipation enhancement by shear flows for generalized diffusion operators in high dimension*, Communications in Mathematical Sciences **23** (2025), 2203–2215, doi:10.4310/CMS.250930031923.

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- D. Coble and S. He, *A note on enhanced dissipation of time-dependent shear flows*, Communications in Mathematical Sciences **22** (2024), 1685–1700, arXiv:2309.15738.
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- D. Albritton, R. Beekie and M. Novack, *Enhanced dissipation and Hörmander's hypoellipticity*, Journal of Functional Analysis **283** (2022), 109522, arXiv:2105.12308.
- S. He, *Enhanced dissipation, hypoellipticity for passive scalar equations with fractional dissipation*, Journal of Functional Analysis **282** (2022), 109319, arXiv:2103.07906.
- M. Coti Zelati and T. D. Drivas, *A stochastic approach to enhanced diffusion*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze **22** (2021), 811–834, arXiv:1911.09995.
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- M. Beck, O. Chaudhary and C. E. Wayne, *Rigorous justification of Taylor dispersion via center manifolds and hypocoercivity*, Archive for Rational Mechanics and Analysis **235** (2020), 1105–1149, arXiv:1804.06916.
- J. Bedrossian and M. Coti Zelati, *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows*, Archive for Rational Mechanics and Analysis **224** (2017), 1161–1204, arXiv:1510.08098.
- M. Beck, O. Chaudhary and C. E. Wayne, *Analysis of enhanced diffusion in Taylor dispersion via a model problem*, Hamiltonian Partial Differential Equations and Applications, Fields Institute Communications **75** (2015), pp. 31–71, arXiv:1501.05877.
- W. Thomson (Lord Kelvin), *Stability of fluid motion: rectilineal motion of viscous fluid between two parallel planes*, Philosophical Magazine (5) **24** (1887), 188–196, doi:10.1080/14786448708628078.

Other smooth flows.

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Rough velocity fields.

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- T. M. Elgindi and K. Liss, *Norm growth, non-uniqueness, and anomalous dissipation in passive scalars*, Archive for Rational Mechanics and Analysis **248** (2024), arXiv:2309.08576.
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3 Stochastic enhanced dissipation

Random velocity fields.

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- J. Bedrossian, A. Blumenthal and S. Punshon-Smith, *Almost-sure enhanced dissipation and uniform-in-diffusivity exponential mixing for advection-diffusion by stochastic Navier-Stokes*, Probability Theory and Related Fields **179** (2021), 777–834, arXiv:1911.01561.

Transport noise.

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4 Deterministic mixing

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5 Stochastic mixing

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